

Lecture 1: Introduction to Linear Programming

Algorithms & Geometry of Linear Programming

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DIMAG Summer School on Combinatorics and Algorithms

August 2026

What is linear programming?

Linear programming (LP): optimize a linear function over finitely many linear inequalities,

$$\max c^T x \text{ s.t. } \mathbf{A}x \leq b.$$

Applications: planning and allocation, network flows, scheduling, game theory, LP relaxations in algorithm design and machine learning, ...

Modern solvers routinely handle instances with millions of variables and constraints.

A brief history

- **1827 — Fourier**: an elimination method for linear inequalities
- **1939 — Kantorovich** (and Koopmans): LP models for economic planning
- **1947 — Dantzig**: the general LP model and the **simplex method**
- **1947–51 — von Neumann; Gale, Kuhn, Tucker**: **duality**
- **1979 — Khachiyan**: polynomial-time solvability (ellipsoid method)
- **1984 — Karmarkar**: **interior point methods**

Today: the model, its geometry, and duality.

Today

The Linear Programming Model

Structure of Linear Programs

Duality: Certifying Optimality

The linear programming model

Linear program (LP)

$$\begin{array}{ll}\max \text{ (or min)} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i \in [m] := \{1, \dots, m\}\end{array}$$

- $\geq$, $=$, and min all reduce to this form: $a_i^T x \geq b_i \iff (-a_i)^T x \leq -b_i$, “ $=$ ” = two inequalities, $\min c^T x = -\max (-c)^T x$
- matrix form: $\max \{c^T x : \mathbf{A}x \leq b\}$, rows of $\mathbf{A} \in \mathbb{R}^{m \times n}$ are $a_1^T, \dots, a_m^T$

x *feasible* if $\mathbf{A}x \leq b$; feasible region $P = \{x : \mathbf{A}x \leq b\}$; $\text{OPT} = \sup\{c^T x : x \in P\}$

conventions: $\text{OPT} = -\infty$ if *infeasible* ($P = \emptyset$), $\text{OPT} = +\infty$ if *unbounded*

Example 1: production planning

Chairs: profit 2, 1 wood, 1 labor. Tables: profit 3, 1 wood, 3 labor.

Available: 4 wood, 6 labor.

$$\begin{array}{ll} \max & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \leq 4 \quad (\text{wood}) \\ & x_1 + 3x_2 \leq 6 \quad (\text{labor}) \\ & x_1, x_2 \geq 0 \end{array}$$

optimal: $(x_1, x_2) = (3, 1)$, value 9

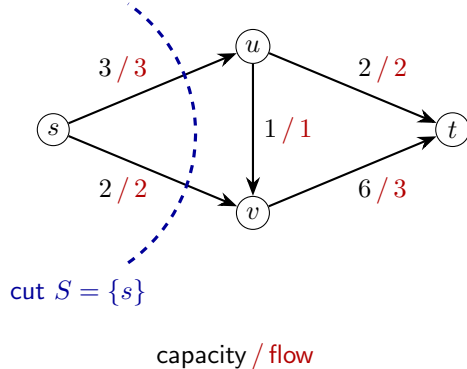
feasibility of $(3, 1)$: easy to check

How to prove optimality?

Answered by duality, at the end of this lecture.

Example 2: maximum flow

Route as much flow as possible from s to t , respecting capacities, conserving flow at internal nodes:



$$\begin{aligned} \max \quad & \sum_{e \text{ out of } s} f_e \\ \text{s.t.} \quad & \sum_{e \text{ into } v} f_e = \sum_{e \text{ out of } v} f_e \quad \forall v \neq s, t \\ & 0 \leq f \leq u \end{aligned}$$

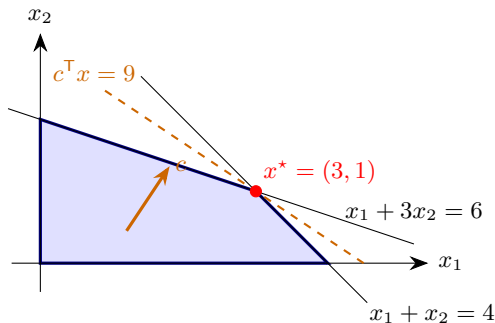
- the flow shown has value 5
- the cut $S = \{s\}$ has capacity $3 + 2 = 5 \Rightarrow$ optimal

The geometry of a linear program

- each constraint $a_i^T x \leq b_i$: a *halfspace*
- feasible region = intersection of m halfspaces: a **polyhedron**
- level sets $\{x : c^T x = \gamma\}$: parallel hyperplanes; slide in direction c until about to leave P

Two observations (in any dimension):

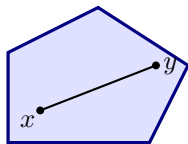
1. the region is **convex**
2. the optimum is attained at a **vertex**



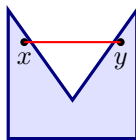
Convexity

Definition

$C \subseteq \mathbb{R}^n$ is *convex* if for all $x, y \in C$, $\lambda \in [0, 1]$: $\lambda x + (1 - \lambda)y \in C$.



convex



not convex

Proposition

Every polyhedron $P = \{x : \mathbf{A}x \leq b\}$ is convex.

Proof. $\mathbf{A}(\lambda x + (1 - \lambda)y) = \lambda \mathbf{A}x + (1 - \lambda)\mathbf{A}y \leq \lambda b + (1 - \lambda)b = b$. □

Local optimality implies global optimality

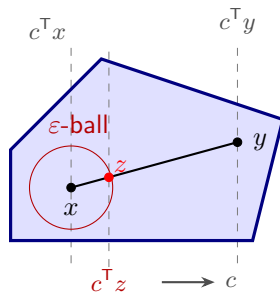
Proposition

If x is a local maximum of $c^T x$ over a convex set P (no better point in P within distance ε), then x is a global maximum.

Better point $y \Rightarrow$ better points on the segment $[x, y]$, arbitrarily close to x .

Along $[x, y]$ the objective climbs steadily from $c^T x$ to $c^T y$: already the point z inside the ε -ball has $c^T z > c^T x$.

Can always make local progress.



Faces of a polyhedron

The boundary of a polyhedron decomposes into flat pieces: vertices, edges, facets, ...

Definition (Face — geometric view)

Convex $F \subseteq P$ is a *face* if: whenever a **strict** average $\lambda x + (1 - \lambda)y \in F$ ($x, y \in P$, $0 < \lambda < 1$), already $x, y \in F$.

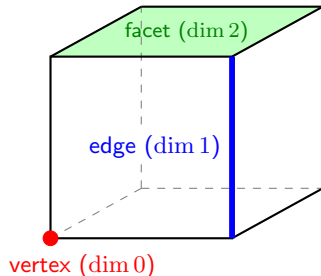
$\dim F :=$ dimension of the affine hull $\text{aff}(F)$. ($P, \emptyset$ are always faces.)

$\dim 0$: *vertices* $\dim 1$: *edges*

$\dim(P) - 1$: *facets*

Cube $[0, 1]^3$, one face highlighted per dimension.

(P itself is its unique $\dim 3$ face.)



Faces = tight constraint sets

For $I \subseteq [m]$: $P_I := \{x \in P : \mathbf{A}_I x = b_I\}$ (set the constraints in I to equality)

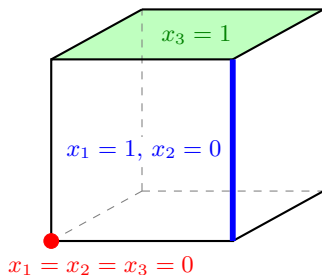
Theorem (Two views of faces)

$\emptyset \neq F \subseteq P$ is a face $\iff F = P_I$ for some $I \subseteq [m]$. For $I = \{i : a_i^\top x = b_i \ \forall x \in F\}$:

$$\text{aff}(F) = \{x \in \mathbb{R}^n : \mathbf{A}_I x = b_I\}, \quad \dim F = n - \text{rank}(\mathbf{A}_I).$$

geometry (faces) $\longleftrightarrow$ linear
algebra (tight rows, rank)

- **facet**: 1 tight row, $\dim = 3 - 1 = 2$
- **edge**: 2 tight rows, $\dim = 3 - 2 = 1$
- **vertex**: 3 tight rows, $\dim = 3 - 3 = 0$

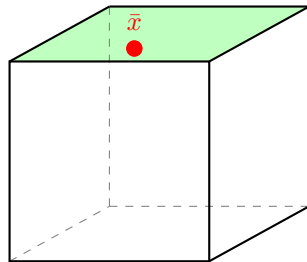


Why $P_I \subseteq F$: reflect through a slack point

$F \neq \emptyset$ a face, $I := \{i : a_i^\top x = b_i \text{ for all } x \in F\}$. ($F \subseteq P_I$ is immediate; the content is $P_I \subseteq F$.)

- pick $\bar{x} \in F$ with $a_j^\top \bar{x} < b_j$ for all $j \notin I$:
pick x^j with slack on inequality $j \notin I$:
 $\bar{x} := \frac{1}{|I|} \sum_{j \notin I} x^j$ is their average
- for $y \in P_I$, **reflect y through $\bar{x}$** :
 $w := \bar{x} + \delta(\bar{x} - y)$
- $\bar{x} = \frac{1}{1+\delta} w + \frac{\delta}{1+\delta} y$ is a **strict average** of $w, y \in P \Rightarrow w, y \in F$

Remark: P_I is the *smallest face containing $\bar{x}$*



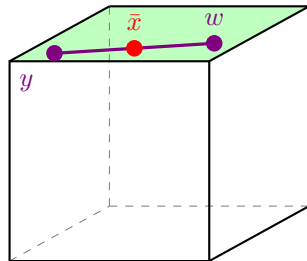
$I = \{x_3 \leq 1\}$: P_I is the top facet

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$I = \{x_3 \leq 1\}$: P_I is the top facet

Polyhedra have few faces

Corollary (Face counting)

A polyhedron with m constraints in $\mathbb{R}^n$ has at most $\binom{m}{n-k}$ faces of dimension k : each k -face is P_J for some set J of $n - k$ linearly independent tight rows.

- **vertices** ($k = 0$): each is the **unique solution** of $\mathbf{A}_J x = b_J$ with $|J| = n$, $\mathbf{A}_J$ invertible $\Rightarrow$ at most $\binom{m}{n}$ vertices
- total: $\sum_k \binom{m}{n-k} \leq 2^m$ faces — finitely many, versus infinitely many points of P

The recession cone

Which directions can one follow forever inside P ?

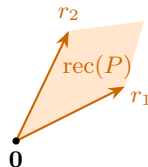
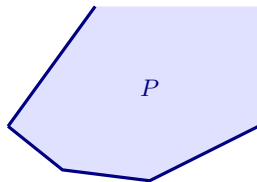
Definition (Recession cone)

$$\text{rec}(P) := \{d \in \mathbb{R}^n : \mathbf{A}d \leq \mathbf{0}\}.$$

For $x \in P$ and $d \in \text{rec}(P)$, the ray $x + td$, $t \geq 0$, stays in P .

Standing assumption: $\text{rank}(\mathbf{A}) = n$.

- P bounded $\iff \text{rec}(P) = \{\mathbf{0}\}$
- edges of $\text{rec}(P)$ are half-lines $[0, \infty) r$ from its unique vertex $\mathbf{0}$: the *extreme rays*
- $V(P) :=$ vertices of P ; $R(P) :=$ one generator r per extreme ray



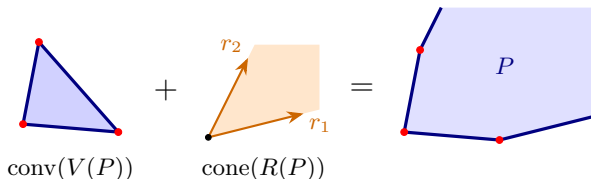
Minkowski-Weyl theorem: the inner description

Theorem (Minkowski-Weyl)

Let $P = \{x : \mathbf{A}x \leq b\} \neq \emptyset$ with $\text{rank}(\mathbf{A}) = n$. Then

$$P = \text{conv}(V(P)) + \text{cone}(R(P)), \quad V(P) \neq \emptyset.$$

Furthermore, every $\text{conv}(V) + \text{cone}(R)$ (V, R finite) is a polyhedron.

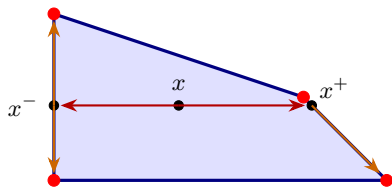


- P bounded: $R(P) = \emptyset$, so $P = \text{conv}(V(P))$ — polytopes are convex hulls of their vertices
- Minkowski (decomposition), Weyl (furthermore). Will show Minkowski direction, Weyl direction in notes.

Proof for bounded P : walking to the vertices

At $x \in P$, let $I = \text{tight constraints}$; induct on $\text{rank}(\mathbf{A}_I)$:

- $\text{rank}(\mathbf{A}_I) = n$: x is a vertex — done
- else pick $d \neq \mathbf{0}$, $\mathbf{A}_I d = \mathbf{0}$; walk both ways along $\pm d$ (keeps I tight); boundedness blocks both walks at $x^\pm$, which gain an independent tight row
- $x \in \text{conv}\{x^-, x^+\}$; recurse on $x^\pm$



$$x \in \text{conv}\{x^-, x^+\} \subseteq \text{conv}(\text{vertices})$$

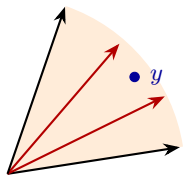
Unbounded P : a walk may continue forever — this happens along an extreme ray of $\text{rec}(P)$, which is how $R(P)$ enters the decomposition. Details in the notes.

Carathéodory's theorem: short representations

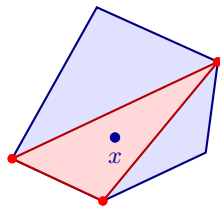
Minkowski's decomposition gives no control on the *number* of generators used.

Theorem (Carathéodory, conic version)

If $y = \mathbf{A}x$ with $x \geq \mathbf{0}$, then $y = \mathbf{A}x'$ for some $x' \geq \mathbf{0}$ whose support columns are linearly independent; in particular $|\text{supp}(x')| \leq \text{rank}(\mathbf{A})$.



in $\mathbb{R}^2$: 2 generators suffice



in $\mathbb{R}^2$: 3 vertices suffice

Consequence: every $x \in P$ uses at most $\dim(P) + 1$ generators in the Minkowski decomposition. Proof: exercise.

The possible outcomes of an LP

Theorem (Trichotomy)

For $\max \{c^T x : \mathbf{A}x \leq b\}$ with $\text{rank}(\mathbf{A}) = n$, exactly one of:

- (a) *infeasible*: $P = \emptyset$
- (b) *feasible and unbounded*: some $r \in R(P)$ has $c^T r > 0$
- (c) *optimal vertex solution*: $\text{OPT} = \max_{v \in V(P)} c^T v$

From the decomposition $x = \sum_v \lambda_v v + \sum_r \mu_r r$:

$$c^T x = \underbrace{\sum_v \lambda_v c^T v}_{\leq \max_v c^T v} + \sum_r \mu_r c^T r \leq \begin{cases} \max_v c^T v, & c^T r \leq 0 \text{ for all } r \in R(P), \\ +\infty, & c^T r > 0 \text{ for some } r \in R(P). \end{cases}$$

A first inefficient algorithm

Brute force over candidate bases and extreme rays

$\forall |J| = \text{rank}(\mathbf{A}_J) = n$, solve $\mathbf{A}_J x = b_J$, test $\mathbf{A}x \leq b$, record $c^T x$ if feasible.

$\forall |J| = \text{rank}(\mathbf{A}_J) = n - 1$, solve $\mathbf{A}_J r = 0$, $r \neq 0$, test if $\pm \mathbf{A}r \leq 0$ and $\pm c^T r > 0$.

Report infeasible / unbounded ray / optimal vertex.

- correct, by the vertex characterization and the trichotomy
- but $\binom{m}{n}$ is **exponential**: for $m = 2n = 100$, over 10^{29} systems

Tomorrow: the **simplex method** — walk from vertex to *adjacent* vertex along improving edges.

How does one prove that no better solution exists?

The production LP: is $(3, 1)$, of value 9, optimal?

$$\max 2x_1 + 3x_2 \text{ s.t. } \underbrace{x_1 + x_2 \leq 4}_{\text{wood}}, \quad \underbrace{x_1 + 3x_2 \leq 6}_{\text{labor}}, \quad x_1, x_2 \geq 0$$

Derive a new inequality: scale constraints by multipliers ≥ 0 and add:

$$\frac{3}{2} \cdot (\text{wood}) + \frac{1}{2} \cdot (\text{labor}) : \quad 2x_1 + 3x_2 \leq \frac{3}{2} \cdot 4 + \frac{1}{2} \cdot 6 = 9$$

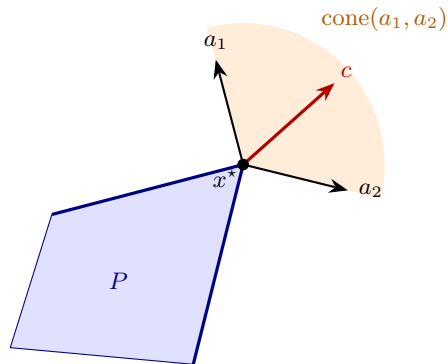
- every feasible point satisfies the derived inequality; its left side is exactly the objective
- with the feasible point $(3, 1)$: **a two-line proof that $\text{OPT} = 9$**

The geometry of optimality certificates

At an optimal vertex x^* , the objective lies in the **cone of the tight constraint normals**:

$$c = y_1 a_1 + y_2 a_2, \quad y_1, y_2 \geq 0$$

The multipliers y are the certificate of the previous slide, supported on tight constraints.



The dual linear program, and weak duality

Primal and dual

$$(P) \quad \max \{c^T x : \mathbf{A}x \leq b\} \qquad (D) \quad \min \{b^T y : \mathbf{A}^T y = c, y \geq \mathbf{0}\}$$

(D) searches for the *cheapest derived upper bound*: multipliers $y \geq \mathbf{0}$ with $\mathbf{A}^T y = c$, bound $b^T y$.

Theorem (Weak duality)

For x feasible in (P) and y feasible in (D): $c^T x \leq b^T y$.

Proof. $c^T x = (\mathbf{A}^T y)^T x = y^T (\mathbf{A}x) \leq y^T b$. □

Any dual feasible y certifies an upper bound. Does the best certificate always match OPT?

Strong duality

Call the primal *bounded* if some $d \in \mathbb{R}$ has $c^T x \leq d$ for every feasible x .

Theorem (Strong duality)

If (P) is feasible and bounded, then (P) and (D) have optimal solutions x^* , y^* , and

$$c^T x^* = b^T y^*.$$

- certification is lossless: every feasible, bounded LP has a matching certificate — flow = cut is one instance of this phenomenon
- boundedness is automatic when both (P) and (D) are feasible (weak duality)
- proof tool: *Fourier–Motzkin elimination* (1827)

Fourier–Motzkin: eliminating one variable

Write $x = (x', x_n)$. Goal: inequalities in x' alone cutting out $\{x' : \exists x_n \text{ with } \mathbf{A}x \leq b\}$.

1. **Sort** the rows by the sign of the coefficient a_{in} of x_n : *lower* ($a_{in} < 0$), *upper* ($a_{in} > 0$), *untouched* ($a_{in} = 0$)

Example. Eliminate x_2 :

$x_1 + 2x_2$	$\leq$	6	upper
$-x_1 + x_2$	$\leq$	1	upper
$-x_2$	$\leq$	0	lower
$-x_1 - x_2$	$\leq$	-1	lower

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2. **Normalize**: divide row i by $|a_{in}|$, so the coefficient of x_n becomes ± 1

Example. Eliminate x_2 :

$$\begin{array}{rclcl} x_1 + 2x_2 & \leq & 6 & \rightsquigarrow & x_2 \leq 3 - \frac{1}{2}x_1 \\ -x_1 + x_2 & \leq & 1 & \rightsquigarrow & x_2 \leq 1 + x_1 \\ -x_2 & \leq & 0 & \rightsquigarrow & x_2 \geq 0 \\ -x_1 - x_2 & \leq & -1 & \rightsquigarrow & x_2 \geq 1 - x_1 \end{array}$$

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3. **Isolate** x_n in the middle:

$$\max_i \ell_i(x') \leq x_n \leq \min_j u_j(x')$$

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$$\max\{0, 1 - x_1\} \leq x_2 \leq \min\{3 - \frac{1}{2}x_1, 1 + x_1\}$$

Fourier–Motzkin: eliminating one variable

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2. **Normalize**: divide row i by $|a_{in}|$, so the coefficient of x_n becomes ± 1
3. **Isolate** x_n in the middle:
$$\max_i \ell_i(x') \leq x_n \leq \min_j u_j(x')$$
4. **Project out** x_n : such an x_n exists $\iff \ell_i(x') \leq u_j(x')$ for every (lower, upper) pair

Example. Eliminate x_2 :

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$$\max\{0, 1 - x_1\} \leq x_2 \leq \min\{3 - \frac{1}{2}x_1, 1 + x_1\}$$

2×2 pairs:

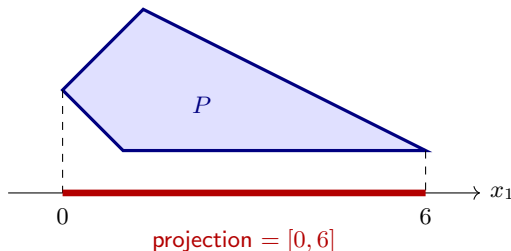
$$-x_1 \leq 0, x_1 \leq 6, -x_1 \leq 1, -x_1 \leq 4,$$

projection = $[0, 6]$ — the last two are redundant

Fourier–Motzkin elimination = projection

Eliminating x_n computes the **projection** of P onto $(x_1, \dots, x_{n-1})$.

- each derived row $\ell_i \leq u_j$ is a **nonnegative combination** of two rows of $\mathbf{A}x \leq b$ (the x_n 's cancel)
- so is any composition of them: eliminating $x_1, \dots, x_k$ leaves rows $(\mathbf{A}^\top y)^\top x \leq b^\top y$ with $y \geq 0$
- cost: the row count can *square* per variable — a proof tool, not an algorithm



Consequence: Weyl

$\text{conv}(V) + \text{cone}(R)$ is polyhedron (notes).

Proof of strong duality (sketch)

Add the objective as a variable t , and consider the system in (x, t) :

$$t - c^T x \leq 0, \quad \mathbf{A}x \leq b \quad (\text{solutions: } x \text{ feasible, } t \leq c^T x)$$

Eliminate $x_1, \dots, x_n$ by Fourier–Motzkin; the surviving inequalities in t read, after normalization,

$$t \leq b^T y^k, \quad \mathbf{A}^T y^k = c, \quad y^k \geq \mathbf{0}, \quad k \in [N].$$

The projection is $T = (-\infty, \min_k b^T y^k]$, so:

- $\tau := \min_k b^T y^k \in T$: some feasible x^* has $c^T x^* = \tau$ (primal optimum, attained)
- the minimizing y^k is dual feasible with value τ (dual optimum, attained)

Full details in the notes.

Duality for general systems: a recipe

Systems in practice mix all constraint types: $\mathbf{A}^u x \leq b^u$, $\mathbf{A}^\ell x \geq b^\ell$, $\mathbf{A}^e x = b^e$.

Goal: certify $c^\top x \leq d$ for every solution.

1. **Orient.** Flip the $\geq$ -rows: $-\mathbf{A}^\ell x \leq -b^\ell$. Equations need no orientation.
2. **Combine.** Multipliers $y^u, y^\ell \geq 0$ on the inequalities, z **unrestricted** on the equations:

$$((\mathbf{A}^u)^\top y^u - (\mathbf{A}^\ell)^\top y^\ell + (\mathbf{A}^e)^\top z)^\top x \leq (b^u)^\top y^u - (b^\ell)^\top y^\ell + (b^e)^\top z$$

3. **Match.** Left-hand coefficients = c , right-hand side $\leq d$.

Strong duality makes the recipe *complete*: for feasible systems, every valid inequality is derivable this way (notes).

The duality cheat sheet (maximization form)

Over the solution set P of the general system, with multipliers indexed by constraint blocks:

Primal–dual pair

$$\begin{array}{ll} \text{(P)} & \max c^\top x \\ & \text{s.t. } x \in P \end{array} \qquad \begin{array}{ll} \text{(D)} & \min (b^u)^\top y^u - (b^\ell)^\top y^\ell + (b^e)^\top z \\ & \text{s.t. } (\mathbf{A}^u)^\top y^u - (\mathbf{A}^\ell)^\top y^\ell + (\mathbf{A}^e)^\top z = c, \quad y^u, y^\ell \geq \mathbf{0} \end{array}$$

- weak and strong duality hold for the pair (feasible + bounded $\Rightarrow$ both optima attained, equal)
- the minimization version is symmetric: orient as $\geq$, maximize the certified lower bound — stated in full in the notes

Certifying infeasibility: the Farkas lemma

Duality certifies optimal values. Can one also certify that $\mathbf{A}x \leq b$ has **no solution**?

Example. $-x \leq -1, x \leq 0$ implies $0 = -x + x \leq -1 + 0 = -1$ (contradiction!)

Theorem (Farkas lemma)

$\mathbf{A}x \leq b$ is infeasible $\iff \exists y \geq \mathbf{0}$ with $\mathbf{A}^\top y = \mathbf{0}$ and $b^\top y = -1$.

- such a y derives $0 \leq -1$ from the system: a one-line proof of inconsistency
- proof: strong duality applied to the phase-one LP $\min \{\mathbf{1}^\top z : \mathbf{A}x - z \leq b, z \geq \mathbf{0}\}$ (minimum total constraint violation) — exercise

Consequence: feasible or infeasible, the correct answer always has a short, checkable certificate: a solution x , or a multiplier vector y .

Summary and what comes next

Today:

- LPs: optimize a linear function over a polyhedron $P = \{x : \mathbf{A}x \leq b\}$
- faces = tight constraint sets; finitely many, with vertices the zero-dimensional case
- Minkowski: $P = \text{conv}(V(P)) + \text{cone}(R(P)) \Rightarrow$ infeasible, unbounded along a ray, or solved at a vertex
- duality: optimal values and infeasibility both admit short certificates — nonnegative combinations of the constraints

Tomorrow: the **simplex method**, and the length of vertex walks (the diameter of polyhedra).

Exercises: convexity of hulls and cones, Carathéodory, complementary slackness, Farkas.